

Red wine analysis

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1. Introduction

The red wine quality dataset, sourced from the UCI Machine Learning Repository, contains 1,599 samples with 11 physicochemical properties and a quality score ranging from 3 to 8, rated by tasters. The objective of this report is to analyze the Red Wine Quality dataset and extract meaningful insights about the factors influencing wine quality by doing PCA and ICA analysis.

This dataset contains **physicochemical properties** (such as acidity, alcohol content, and sulfur dioxide levels) and a **quality rating** given by wine tasters.

2. Data Loading and Preprocessing

Red Wine Quality Dataset

fixed.aci dity	volatile.aci dity	citric.a cid	residual.s ugar	chlori des	free.sulfur.di oxide	total.sulfur.di oxide	densi ty	pH	sulpha tes	alcoh ol	quali ty
7.4	0.70	0.00	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5
7.8	0.88	0.00	2.6	0.098	25	67	0.9968	3.20	0.68	9.8	5
7.8	0.76	0.04	2.3	0.092	15	54	0.9970	3.26	0.65	9.8	5
11.2	0.28	0.56	1.9	0.075	17	60	0.9980	3.16	0.58	9.8	6
7.4	0.70	0.00	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5

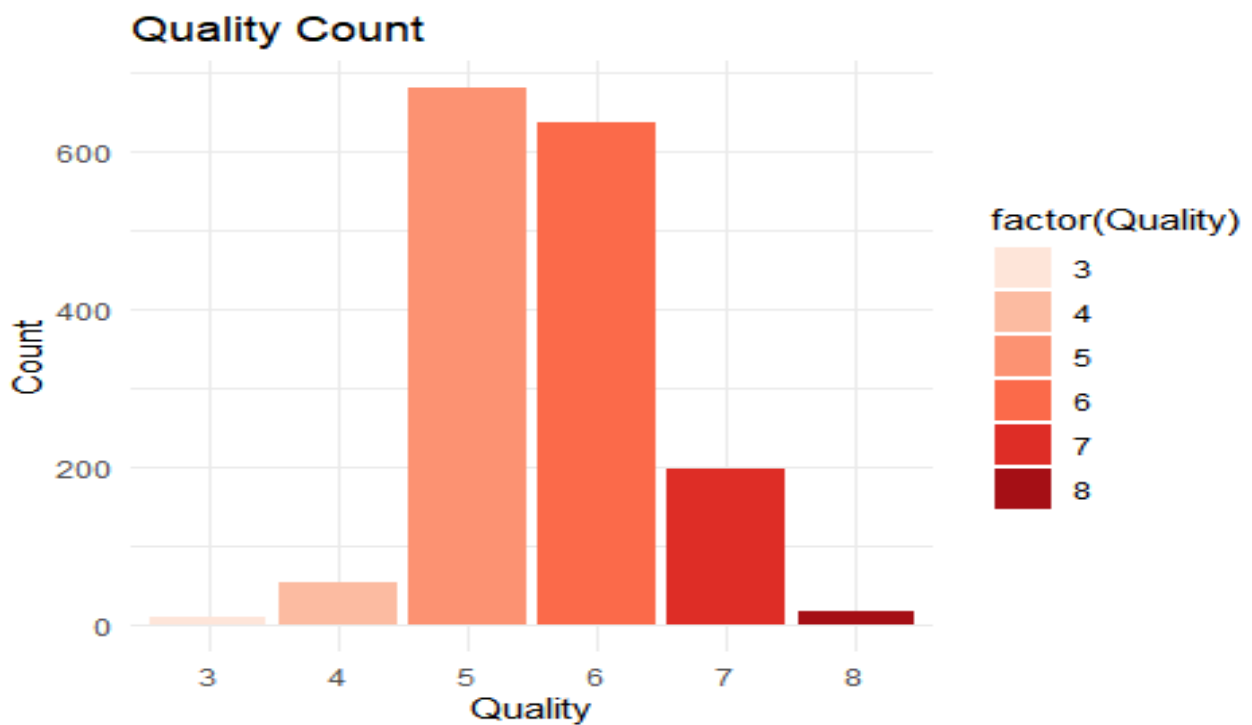
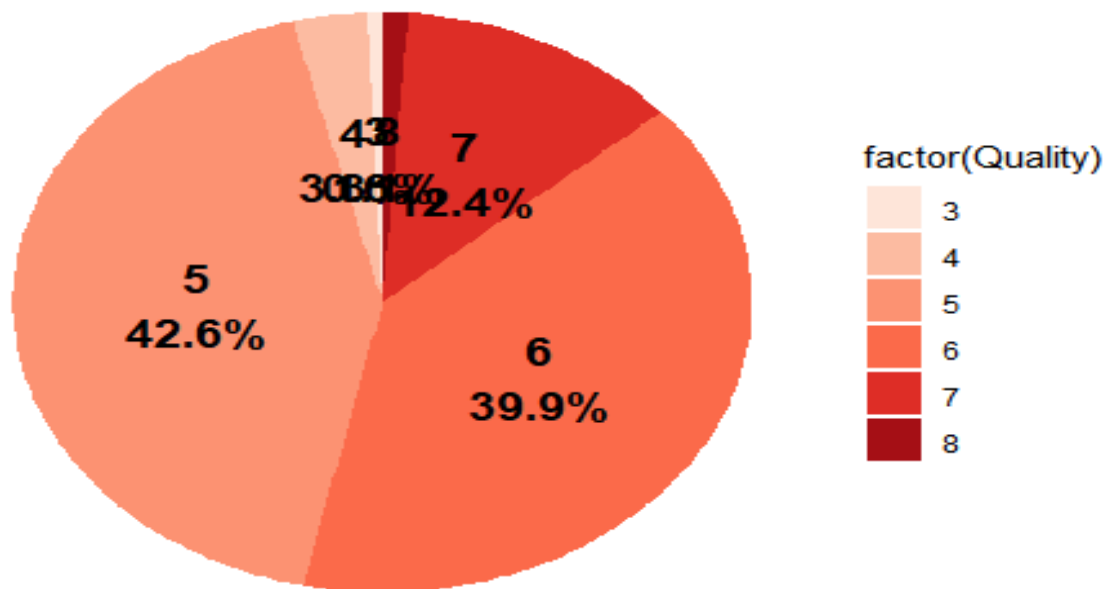
- No missing data simplifies analysis, ensuring reliability.

3. Exploratory Data Analysis (EDA)

3.1 Distribution of Wine Quality Scores

We visualize the frequency of quality scores to assess distribution.

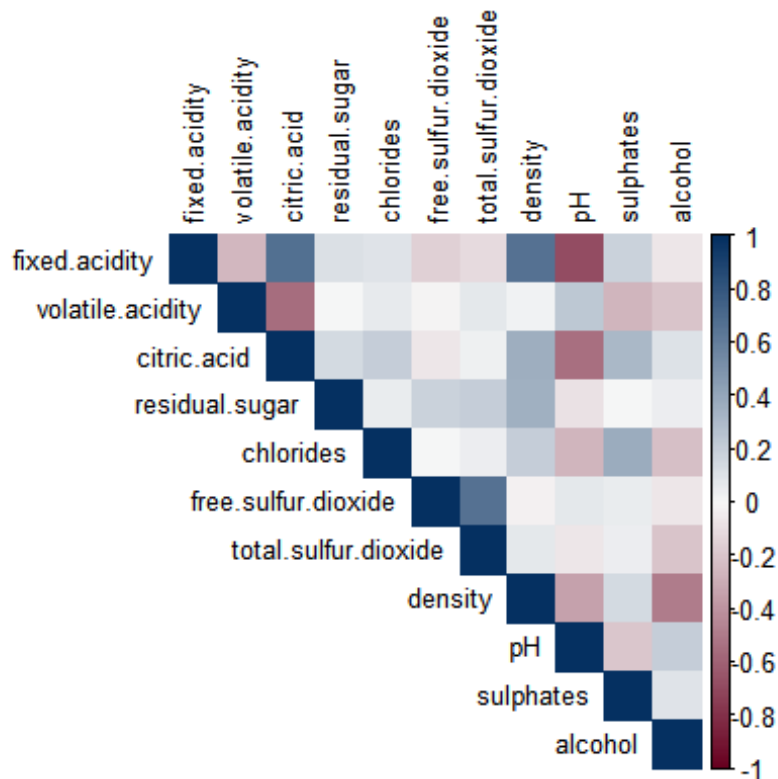
Quality Distribution



The distribution is imbalanced, with most wines average (5–6). This could mean exceptional wines (7–8) have unique traits, but their rarity might limit statistical power, affecting model performance.

3.2 Correlation Matrix

We examine feature correlations to identify relationships.



- Strong correlations exist: fixed.acidity ties to density (denser wines are more acidic), and alcohol has a mild positive link to quality (~0.47), suggesting it boosts quality. These relationships guide PCA's focus, highlighting interdependencies like acidity-density links.

3.3 Multiple Histograms for All Features

We plot histograms to understand feature distributions.

istogram of fixed.istogram of volatile.Histogram of citric.stogram of residual



Histogram of chloridistogram of free.sulfungram of total.sulfur Histogram of dens



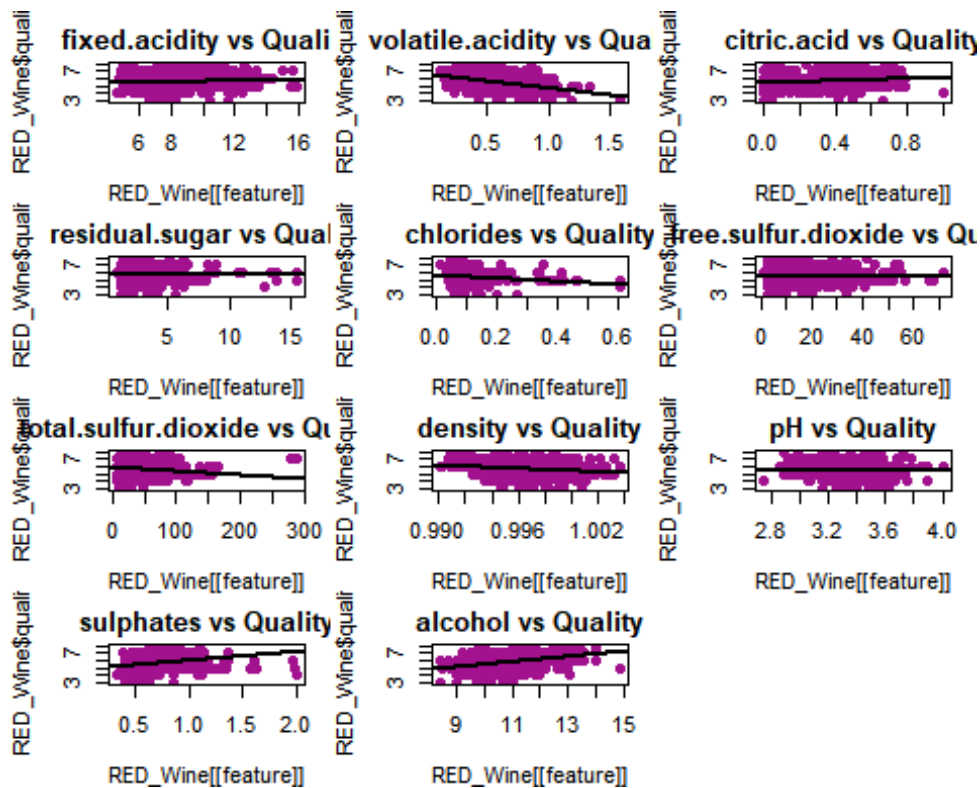
Histogram of pHHistogram of sulphHistogram of alcol Histogram of qual



- Normal distributions (e.g., alcohol) suggest consistent properties, while skewed ones (e.g., chlorides) indicate outliers—wines with unusually high saltiness. This shapes PCA's variance capture, as skewed features may dominate.

3.4 Scatterplots of Features vs. Quality

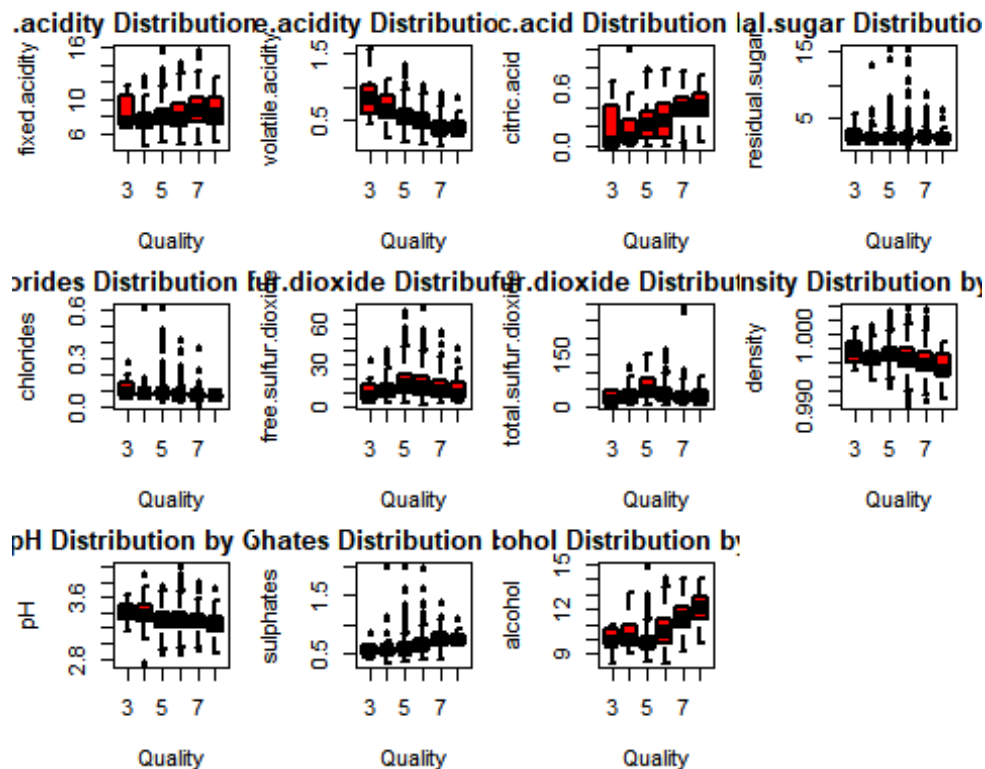
We plot each feature against quality to identify trends.



- No single feature screams “quality predictor.” alcohol’s weak trend suggests it helps, but quality likely stems from a mix of factors—PCA and ICA will dig deeper, revealing combined effects.

3.5 Boxplots by Quality

We compare feature distributions across quality levels.



- Subtle shifts emerge—higher alcohol and lower volatile.acidity tie to better wines (7–8). Overlaps in boxes show these aren’t strict rules, just hints, suggesting quality is a blend of factors.

4. Principal Component Analysis (PCA)

PCA reduces dimensionality by identifying directions of maximum variance, helping simplify the data.

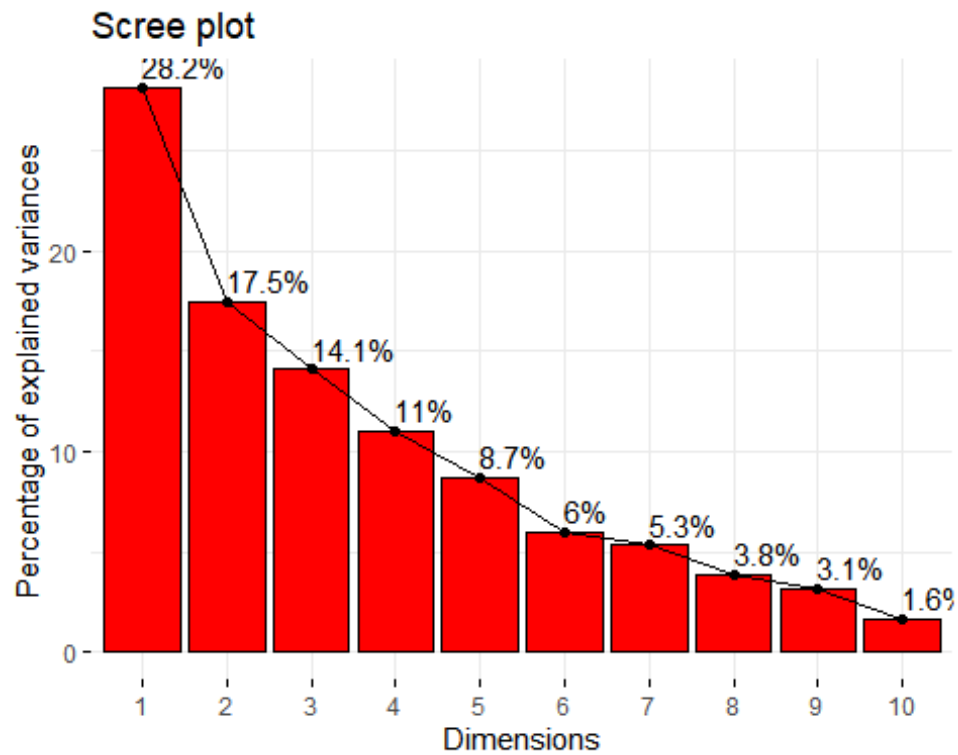
4.1 Setup and Standardization

We standardize the data to ensure equal contribution.

```
wine_scaled <- scale(RED_Wine %>% select(-quality))
pca_result <- prcomp(wine_scaled)
```

4.2 Scree Plot

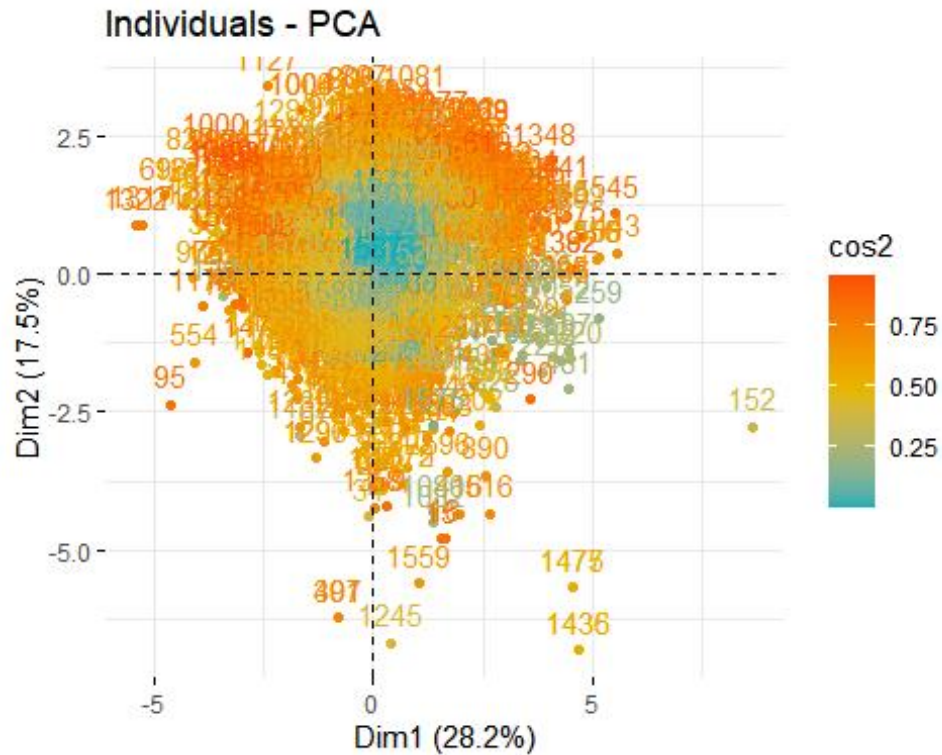
We visualize variance explained by each component.



Comments: - Nearly half the variation is in the first two PCs—good, but not enough to fully explain quality. PC3 onward adds detail incrementally, suggesting we might need more components for a complete picture.

4.3 Individual Factor Map

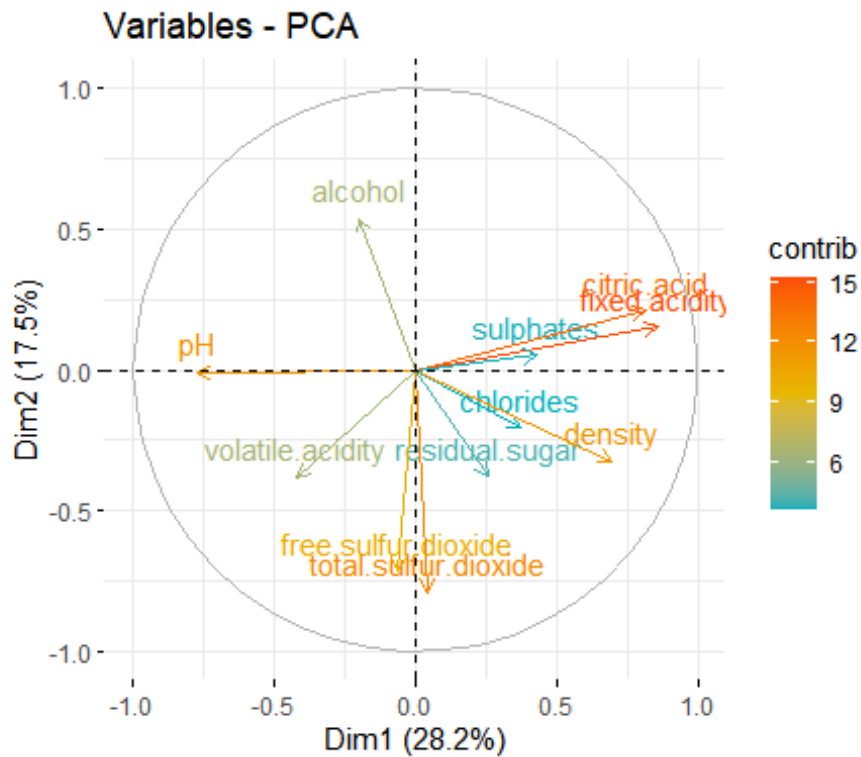
We plot wines on PC1 vs. PC2, colored by fit.



Comments: - Wines spread in a cloud—PC1 (28.2%) and PC2 (17.5%) explain 45.7% of variation. No clear groups mean these components capture broad differences, not quality-specific ones, aligning with.

4.4 Variable Contributions

We examine feature influence on PCs.

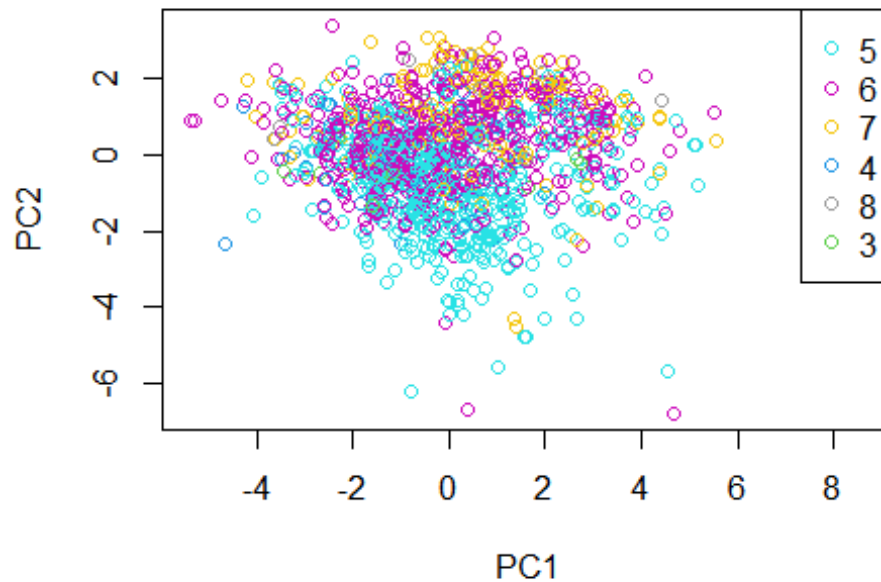


Comments: - Acidity and density dominate PC1, alcohol and sulphates PC2. These are the “big levers” of wine variation—key for reporting, as they align with chemical properties affecting taste and structure.

4.5 PCA by Quality

We color by quality to check separation.

PCA of Red Wine Dataset



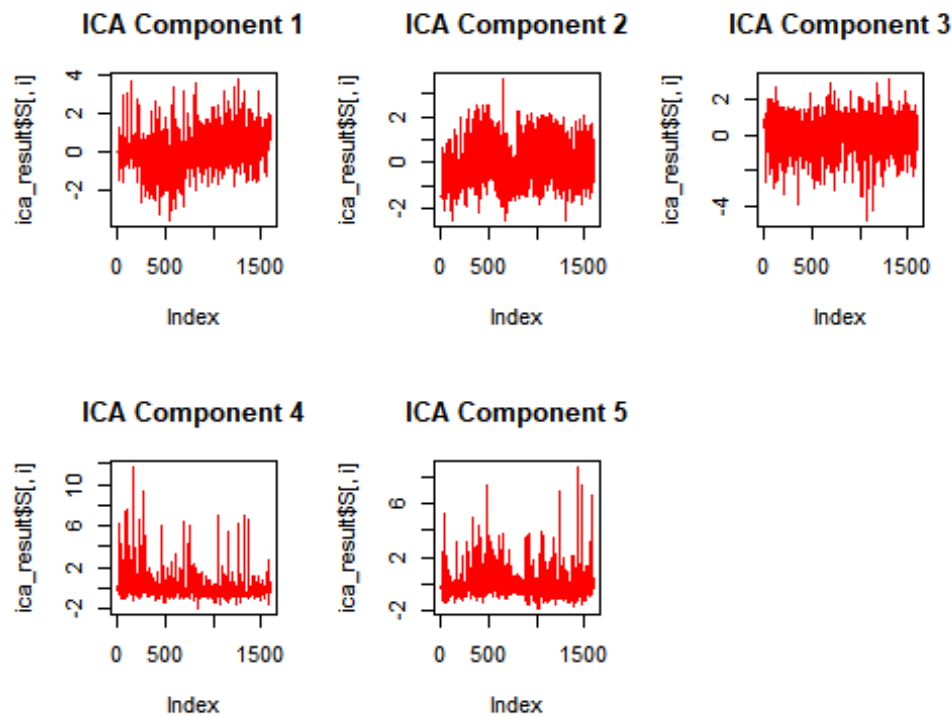
Comments: - Quality doesn't separate in PC1/PC2—overlaps suggest quality isn't just about these main differences, indicating a need for deeper analysis.

5. Independent Component Analysis (ICA)

ICA seeks statistically independent components, potentially revealing hidden factors.

5.1 Visualizing ICA Components

We plot each component's values.

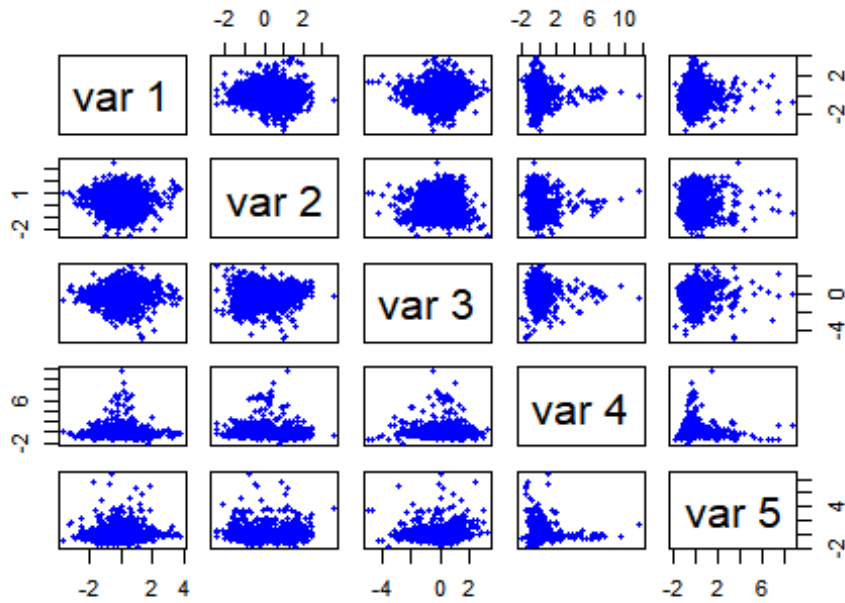


Comments: - Spikes flag wines with unique profiles—maybe high acidity or sulfur. Each component captures a distinct “flavor,” suggesting multiple independent chemical signatures, potentially tied to quality nuances.

5.2 Scatterplot of ICA Components

We check relationships between components.

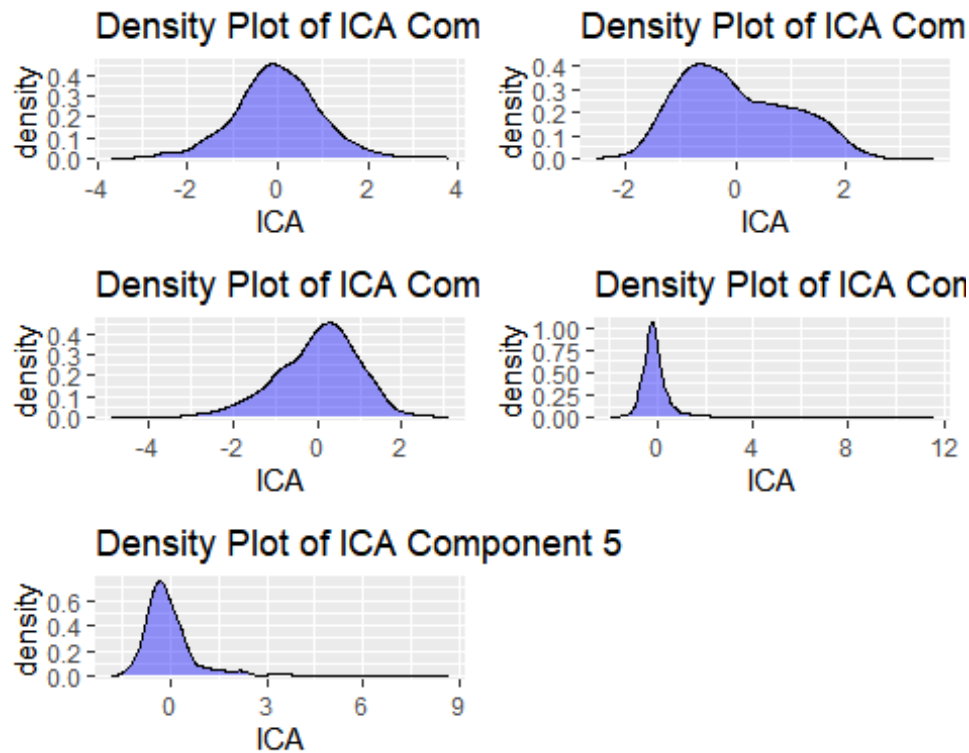
Scatterplot of ICA Components



Comments: - No correlations confirm independence—ICA isolates unique signals, unlike PCA's blended view, offering a fresh lens on wine chemistry.

5.3 ICA Density Plot

We plot the distribution of each component.



Comments: - Non-Gaussian shapes are ICA's strength, highlighting rare traits (e.g., extreme sulphates) that might tie to quality, providing insights into outliers or unique wine profiles.

EDA Insights:

Quality is imbalanced (mostly 5–6, rare 7–8), suggesting exceptional wines have unique traits. Alcohol (~0.47 correlation) boosts quality; volatile acidity (negative correlation) detracts.

PCA Findings:

PC1 (acidity, density, citric acid) and PC2 (alcohol, sulphates) explain ~45% of chemical variation, but quality scores don't cluster, indicating quality isn't a simple function of these axes.

ICA Findings:

Independent components might isolate factors like fermentation (affecting acidity) or preservatives (sulphates/alcohol), with potential quality links (e.g., a component favoring low volatile acidity and high alcohol aligning with quality 7–8). Implications for Red Wine Quality

In this red wine quality analysis: The dataset's variation is dominated by acidity-density (PC1) and alcohol-sulphates (PC2), critical to wine's structure and taste. However, these

don't fully explain quality, suggesting sensory quality arises from subtle interactions. Quality Complexity: Higher-quality wines (7–8) likely combine higher alcohol (enhancing body) and lower volatile acidity (reducing off-flavors), but PCA shows no single “quality formula.” ICA hints at independent processes (e.g., fermentation control) that might differentiate exceptional wines. ICA's non-Gaussian components could identify unique wines, guiding quality control or innovation.

Conclusion

For the red wine quality dataset, PCA reveals that while acidity, density, alcohol, and sulphates drive chemical diversity, they don't directly dictate quality, pointing to a nuanced interplay of factors. ICA complements this by suggesting independent chemical signatures—potentially tied to winemaking techniques—that might better align with quality differences. For instance, a component emphasizing low volatile acidity and high alcohol could mark higher-quality wines, offering a path to enhance production. Future steps could use these components in predictive models or cluster wines to pinpoint quality subgroups, ultimately aiding producers in crafting superior red wines.